Whole-Number Exponent Laws

Concept

Exponent laws simplify repeated factors. They do not combine unlike terms in sums.

Product / quotient

$a^m a^n = a^{m+n}$; $a^m / a^n = a^{m-n}$ for $a \neq 0$.

Power of a power

$(a^m)^n = a^{mn}$.

Product power

$(ab)^n = a^n b^n$; $a^0 = 1$ for $a \neq 0$.

Warning

$(a + b)^2$ is not $a^2 + b^2$. Grade 9 uses whole-number outcomes.

Example 1

Simplify $(2a^3b)^2$.

Solution

1. Square every factor: $2^2 (a^3)^2 b^2$.
2. Result: $4a^6b^2$.

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Example 2

Simplify $(18x^5y^3)/(6x^2y)$.

Solution

1. State restrictions: $x \neq 0$ and $y \neq 0$.

2. Divide coefficients and subtract exponents: $3x^3y^2$.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Simplify m^3m^5.

2. Simplify $(3p^2)^3$.

3. Simplify $20a^6/(5a^2)$, stating restrictions.

4. Evaluate $(-3)^2 + 2^0$.

5. Is $(x + 2)^2 = x^2 + 4$? Disprove with a value.

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Practice answers | check after trying

1. m^8.

2. $27p^6$.

3. $4a^4$, $a \neq 0$.

4. $9 + 1 = 10$.

5. No: $x = 1$ gives 9 on the left and 5 on the right. The missing term is $4x$.